

BHAGYASHRI PAGAR

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Education

Northeastern University

Sep 2024 - May 2026

Master of Science in Information Systems (GPA: 3.85)

Boston, MA

Relevant Coursework: Data Architecture for Business Intelligence, Data Science, Database Management & Design, Application Engg

Pune University

Aug 2019 - May 2023

Bachelor of Engineering in Information Technology (GPA: 3.98)

Pune, India

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management, Object-Oriented Programming, Artificial Intelligence, NLP, Cloud Computing, Machine Learning, Big Data Analytics, Data Science, Deep Learning

Technical Skills

Languages	Java, Python, C++, Kotlin, R, JavaScript, Go, PHP, C#, SQL
Databases	MySQL, MongoDB, PostgreSQL, Snowflake, MS SQL Server, BigQuery, Databricks
Frameworks & Libraries	React, NodeJS, ExpressJS, AJAX, Spring Boot, Angular, FastAPI, HTML5, CSS, TypeScript, Django
Machine Learning & AI	Pandas, NumPy, PyTorch, Matplotlib, Seaborn, TensorFlow, Keras, Scikit-learn, OpenCV, NLTK
DevOps & Cloud	AWS, Azure, Google Cloud (GCP), Linux/Unix, Docker, Kubernetes, Terraform, Jenkins, Git, Firebase
Tools	Alteryx, Airflow, Jupyter, Visual Studio, DBeaver, Figma, Excel, Power BI, Tableau, Android Studio

Professional Experience

Sunanda Infotech

Jul 2023 – Jul 2024

Software Engineer

Nashik, India

- Developed and deployed full-stack web applications using **React.js**, **TypeScript**, **Node.js**, and **MongoDB**, implementing responsive design and **REST APIs** that improved web performance by **35%** and supported scalable production environments
- Architected scalable microservices with **Java Spring Boot** and **MySQL**, containerized via **Docker**, and built reusable middleware that reduced API response times by **40%** while enhancing maintainability and fault isolation
- Configured secure authentication and access control using **JWT**, **OAuth 2.0**, and role-based policies, integrated **AWS S3**, **Firebase APIs**, and web services, improving system reliability by **25%** and ensuring compliance with security standards
- Led and collaborated in Agile sprints with Git-based version control and daily standups, boosting sprint velocity by **30%**; documented APIs, workflows, and data models in **GitHub** to streamline cross-functional collaboration

The Sparks Foundation

Apr 2022 – May 2022

Web Development Intern

Pune, India

- Devised a responsive website using **HTML**, **CSS**, & **JavaScript**, integrating a secure payment gateway following best practices in SDLC & quality assurance that enabled multi-platform online transactions & accelerated site traffic by **25%**
- Delivered user interface design/ user experience(UI/UX) & performance best practices, upgraded customer experience & retention by **15%** & elevated page load speed by **28%** through refined DOM manipulation, image compression, & accessibility enhancements

Oasis Infobyte

Jan 2022 – Mar 2022

Web Development Intern

Pune, India

- Engineered a cross-platform food delivery web app using **React.js**, **TypeScript** frontend, & **Node.js** backend, implementing component-based architecture, **Redux** for state management, & **RESTful APIs** that increased user engagement by **20%** & improved performance
- Enhanced the frontend with code splitting & modular architecture using, reducing load times by **30%** & improving retention on low-bandwidth networks; implemented debugging and error-handling practices to ensure reliable production support

Projects

Multi-Agent Financial Research Platform | LangGraph, FastAPI, Snowflake, Pinecone, React.js, Vector Embeddings

Mar 2025-May 2025

- Prototyped a full-stack AI-driven financial markets research platform integrating **LangGraph** agents with a **FastAPI** backend & **React.js** frontend to deliver real-time semantic search across **10GB+** financial documents
- Integrated **pinecone** for vector-based retrieval & **PostgreSQL** compatible **Snowflake** for data warehousing, achieved **95%+** result relevance & reduced analyst research time by **70%** through contextual multi-agent reasoning & implemented ETL workflows

TourEase-Custom Travel Package & Vendor Integration Platform | React.js, Node.js, SQL Server, T-SQL, PowerBI

Jan 2025- Feb 2025

- Built a full-stack travel booking platform with an entrepreneurial mindset using **React.js** & **Node.js**, load-tested for **1,000+** concurrent users & embedded 4 vendor service modules with **RESTful APIs** & microservices architecture for modular deployment
- Designed & normalized a **20+** table schema in SQL Server database with stored procedures, triggers, & optimized indexing, achieving up to **45%** faster query performance in test loads; visualized tour metrics using **PowerBI dashboards** for data visualization

LightRAG - Retrieval-Augmented Generation System | Python, PyTorch, FAISS, Google Gemini API

Sep 2024- Nov 2024

- Implemented a low-latency high-performance, lightning-fast RAG system with dual-level retrieval (entity + concept) & graph-based indexing using **Python**, **PyTorch**, & **FAISS**, incorporating dual-level entity & concept retrieval, boosting semantic match accuracy by **23%** & reducing end-to-end latency by **83%**
- Utilized **Google Gemini API** and **LangChain** for LLM-driven summarization & quality assurance, enabling context-rich, low-latency responses across diverse inputs, reduced document fragmentation by **96.9%** using adaptive chunking, improving retrieval precision